FLOOD FORCASTING

USING MACHINE LEARNING

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INTRODUCTION

UNDERSTANDING THE NEED FOR FLOOD PREVENTION

- Floods are among the most devastating natural disasters, causing loss of life, infrastructure damage, and economic disruption worldwide.
- Traditional flood prevention methods, such as dams and levees, are no longer sufficient on their own.
- With the rise of Artificial Intelligence (AI) and Machine Learning (ML), data-driven models can now learn flood patterns directly from historical rainfall and river discharge data.

CAUSES OF FLOODING

Floods occur due to a combination of natural and human-induced factors, often leading to devastating consequences. Understanding these causes is essential for developing effective flood prevention strategies.

- Heavy Rainfall & Storms
- River Channel Overflow
- Upstream Runoff
- Deforestation And Climate Change

PROBLEM STATEMENT

Predict river water level or discharge in advance to minimize flood damage using publicly available hydrological data.

PROPOSED SOLUTION

Step 1: **Data Collection**

- Gather historical data such as rainfall, river discharge, temperature, soil moisture, and water level from reliable sources (meteorological and hydrological departments).

Step 2: **Data Preprocessing**

- Handle missing values, remove outliers, normalize data, and align time-series records for consistency.

Step 3: **Feature Selection**

- Identify key parameters influencing floods — rainfall intensity, upstream discharge, soil saturation, etc.

Step 4: **Model Development**

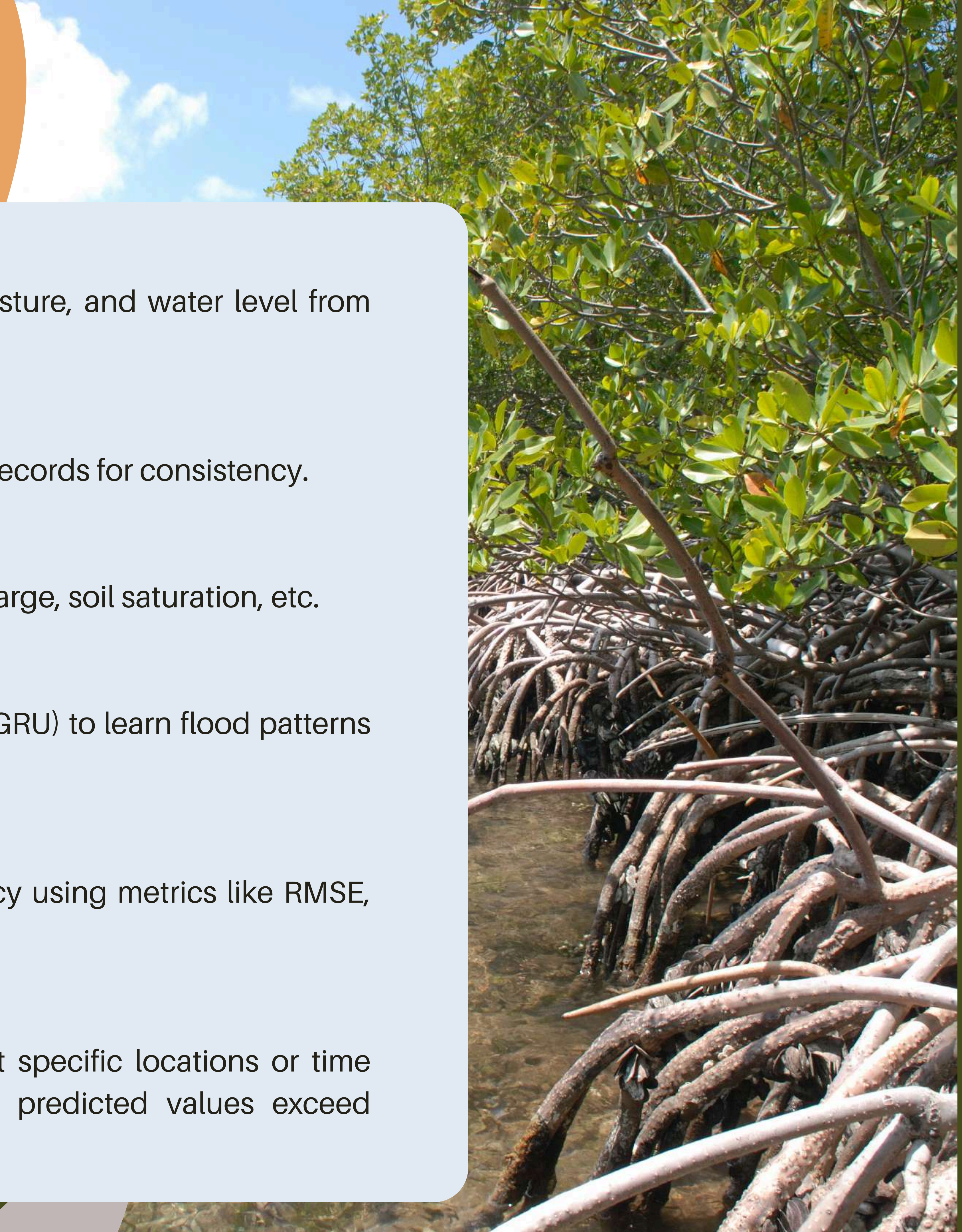
- Apply machine learning and deep learning algorithms (e.g., Random Forest, GRU) to learn flood patterns from past data.

Step 5: **Model Training and Validation**

- Split dataset into training and testing subsets to evaluate prediction accuracy using metrics like RMSE, MAE, and R^2.

Step 6: **Flood Prediction**

- Use the trained model to forecast future water levels or flood likelihood at specific locations or time intervals through dashboards or maps and trigger early warnings when predicted values exceed threshold limits.



ML PIPELINE

START
Step 1 → Merge USGS + NASA POWER on daily date index
Step 2 → Engineer 6 lag/rolling features + create 7-day-ahead target
Step 3 → Chronological 80/20 Train-Test Split Train: 13,100 days (1981–2016) Test: 3,275 days (≈2017–present) No shuffling → prevents data leakage
Step 4 → Train 3 models: Persistence Baseline (today's Q = Q+7) Random Forest — 300 trees, parallel Gradient Boosting — 500 estimators, lr=0.05
Step 5 → Evaluate with MAE and RMSE on test set
END

DATA SOURCES

USGS + NASA POWER — 1981 TO PRESENT

- USGS National Water Information System (NWIS)
  - St. Louis, MO gauge station → Target discharge (cfs)
  - Keokuk, IA gauge station → Upstream predictor
  - Grafton, IL + Thebes, IL → Additional upstream stations
  - 16,388 daily records · 44 years of data
- NASA POWER Meteorological API
  - Daily Precipitation (PRECTOT, mm/day)
  - Temperature — Mean (T2M) and Max (T2M_MAX) in °C
  - Relative Humidity (RH2M, %)
  - Wind Speed (WS2M, m/s)
  - Surface Pressure (PS, kPa)

FEATURE ENGINEERING

TIME-AWARE FEATURES BUILT FROM RAW DATA

Rainfall Accumulation:	Discharge Lag Features:
rain_3day → 3-day rolling sum of PRECTOT	stl_lag1, stl_lag3 → St. Louis discharge 1 and 3 days back
rain_7day → 7-day rolling sum of PRECTOT	keokuk_lag1, keokuk_lag3 → Upstream Keokuk 1 and 3 days back
These capture antecedent soil wetness and storm accumulation	Target: discharge_future = St. Louis discharge shifted 7 days forward

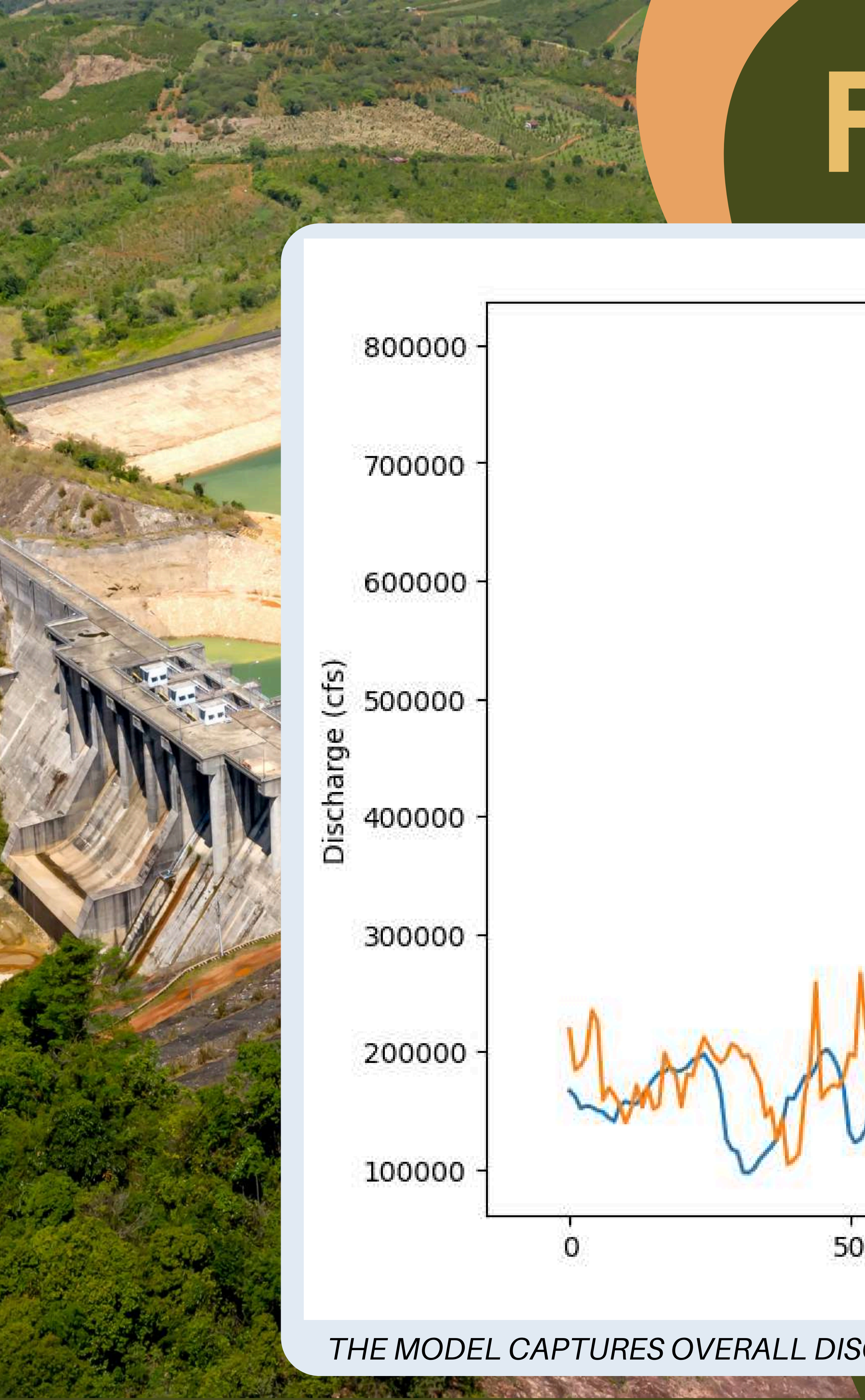
MODEL PERFORMANCE COMPARISON

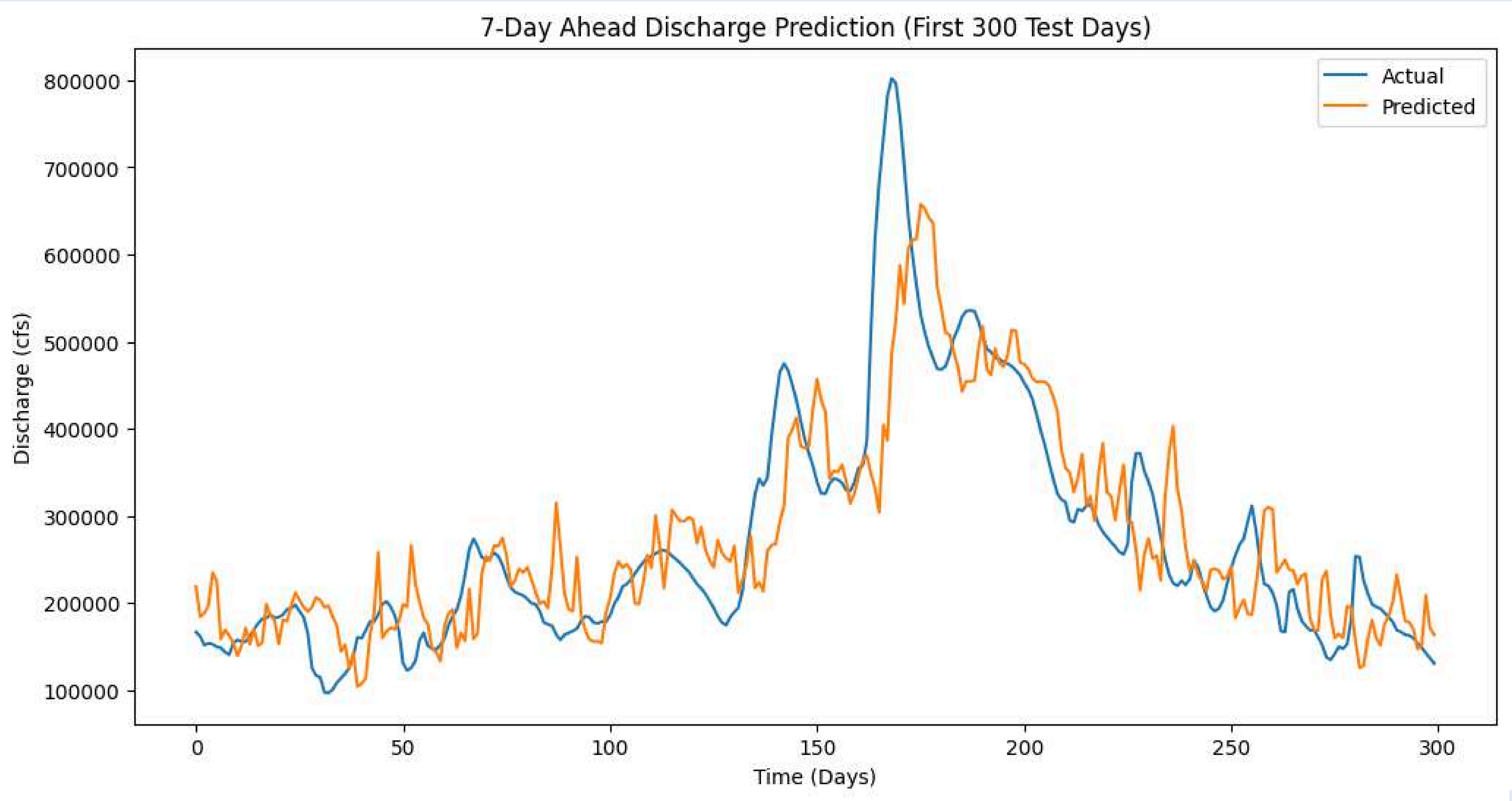
Model	MAE (cfs)	RMSE (cfs)	Relative Performance
Persistence (Baseline)	43,408	64,447	Reference
Random Forest	46,570	66,003	Slightly worse
Gradient Boosting	42,604	61,764	**Best (+4.2%)**

GRADIENT BOOSTING IMPROVED FORECASTING ACCURACY BY APPROXIMATELY 4–5% OVER THE PERSISTENCE BASELINE.

THE STRONG PERFORMANCE OF THE PERSISTENCE BASELINE REFLECTS HIGH TEMPORAL AUTOCORRELATION IN RIVER DISCHARGE.

FORECAST VISUALIZATION



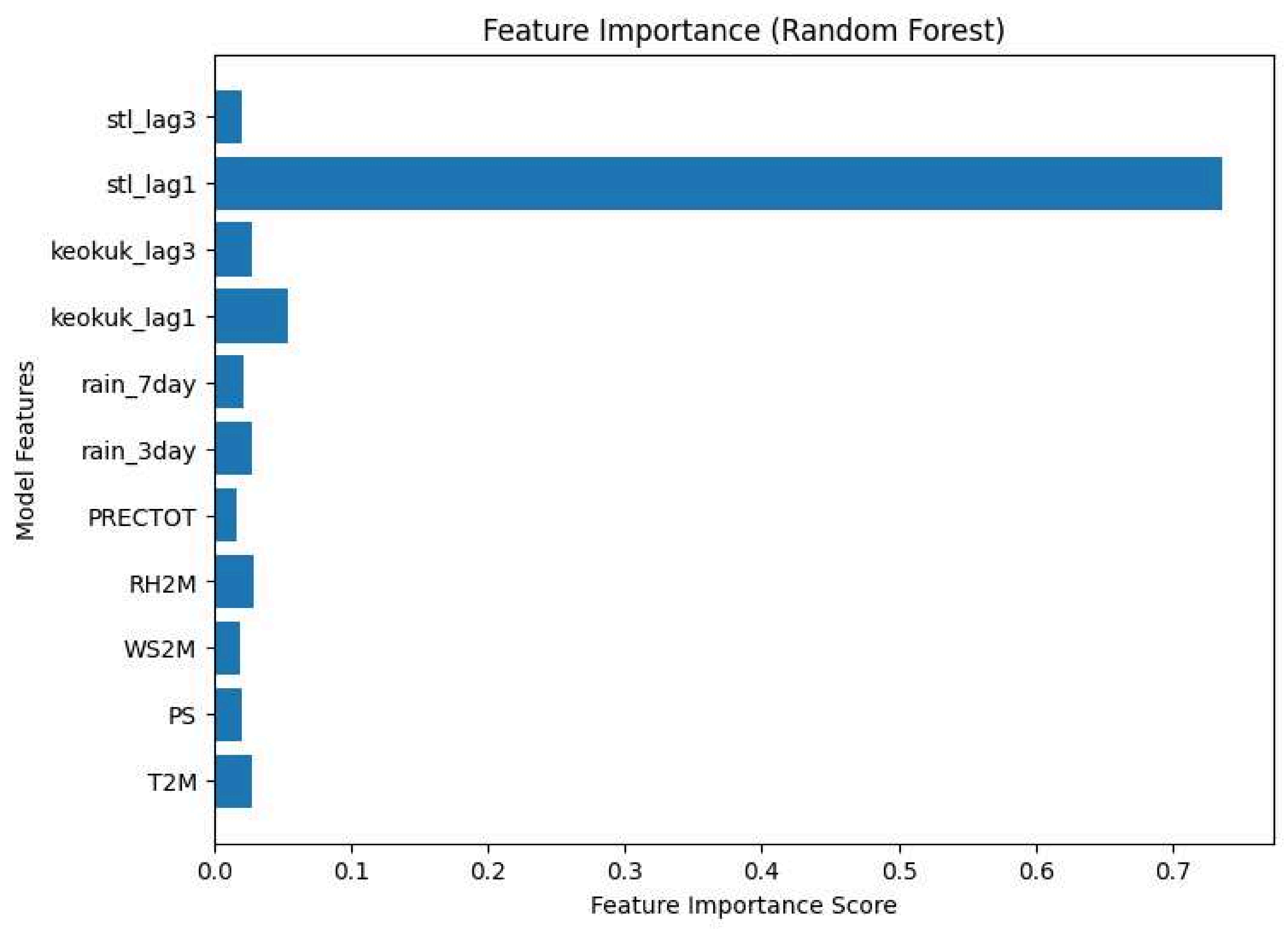


THE MODEL CAPTURES OVERALL DISCHARGE TRENDS AND MAJOR FLOOD EVENTS BUT SLIGHTLY UNDERESTIMATES EXTREME PEAKS.

FEATURE IMPORTANCE

- stl_lag1 dominates (~74%)
- Upstream flow (Keokuk) is second most important
- Rainfall variables contribute but are secondary

Short-term discharge forecasting is strongly driven by hydraulic persistence and upstream propagation.



CONCLUSION

- Gradient Boosting achieved the best RMSE (61,550 cfs) — a 4-5% improvement over the naive persistence baseline
- Random Forest provides interpretable feature importance: stl_lag1 dominates at ~74%, confirming strong river autocorrelation
- Upstream Keokuk flow is the 2nd most important predictor — validating multi-station data collection
- Rainfall features provide secondary but meaningful signal — allowing ML to beat the baseline
- Extreme flood peaks are slightly underestimated due to their rarity in training data

THANK YOU.